

Cyber-Physical Co-Simulation for Smart Grid Security

Impact of Cyber-Induced False Data Injection on Solar-Rich Distribution Networks

Problem Statement

The Gap

Power grids are rapidly integrating with solar panels, relying on smart inverters for voltage regulation without adequate cybersecurity measures.

The Threat

Cyber-attackers can inject False Data (FDI) into solar inverter controllers, forcing voltage instability that leads to blackouts or equipment damage.

The Need

A lightweight co-simulation tool to measure cyber-attack thresholds for grid destabilization

Current Research Challenges

Cross-Layer Gaps

Existing research treats "Cyber" and "Physical" layers separately, missing critical interdependencies.

Scalability Issues

Current frameworks struggle with dynamic behaviors in large-scale interconnected systems.

Real-Time Adaptation

Lack of real-time adaptability to evolving threats and complex external factors.

Data Scarcity

Limited realistic datasets due to privacy concerns and vulnerability exposure risks.

Cost Dependencies

High-fidelity simulations rely on expensive proprietary hardware and software.

Integration Complexity

Inconsistent modeling of threat propagation across physical, network, and application layers.

Key Research References

Tool-Focused Studies (2024-2025)

Lanzinger et al. (2025) - Co-Simulation for Cyber-Attack Assessment

arXiv: 2410.10430 (2024) - Cyber-Physical Energy Systems

Attack Modeling & AI (2024-2026)

Dec 2024 - Multi-Stage Attack-Defense Framework

Jan 2026 - Intrusion Detection Using Inception-V4

Power System Integration (2025)

Aug 2025 - PV Penetration Impact & Pandapower

Technology Stack

Power Simulation tool

pandapower

Python-based power flow solver

Combines pandas data analysis with PYPOWER solver for distribution network analysis

Co-Simulation tool

HELICS

Backbone Coordination simulator

Synchronizes power and network data across simulation domains

Network simulation tool

ns-3

Network simulator

Simulates communication delays, packet drops, and network congestion

Analysis & Visualization

Matplotlib / Seaborn

Python plotting libraries

Voltage violation curves and attack impact visualization

Experimental Setup

Physical Grid (pandapower)

- IEEE 13/34-node distribution feeder
- Multiple solar PV systems (10-50 kW)
- VDE standard compliant

Cyber Layer (ns-3)

- MQTT communication simulation
- Network delays and packet loss
- Attack injection capabilities

Integration (HELICS)

- Synchronizes physical and cyber domains
- Real-time co-simulation
- Enables cyber-physical attack testing

Evaluation Metrics

Voltage Stability Index - Deviation of **voltage** from the safe operating range ($230V \pm 10\%$)

Detection Latency - How fast the system detects an ongoing attack

Network Impact - How communication delays and packet loss affect attack behaviour

Our Research vs. Existing Approaches

Existing Research

- Expensive proprietary tools
- Complex, inaccessible
- Separate domains

Our Approach

- Open-source Python framework
- European distribution grids
- solar PV integration

Key Innovation

EU solar-rich distribution networks with integrated security testing

4-Week Project Timeline

1

Environment Setup & Baseline

Install Python, HELICS, pandapower, ns-3. Setup IEEE feeder in pandapower. Run baseline power flow simulation.

2

Integration & Communication

Integrate HELICS and ns-3 to create communication link. Implement MQTT protocol simulation. Add solar PV systems to network.

3

Attack Implementation & Testing

Launch FDI and Replay attack scripts. Record voltage deviations and system responses. Test multiple attack intensities.

4

Analysis & Documentation

Visualize attack impacts with Matplotlib. Generate synthetic datasets (optional).

Expected Outcomes & Impact

Attack-Impact Model

Validated model showing which network nodes are most vulnerable to FDI attacks based on solar penetration levels

Synthetic Datasets (currently optional but can be implemented)

Comprehensive datasets for training AI-based anomaly detection systems without exposing real grid data

Functional Co-Simulation Testbed

Cost-Effective

Fully operational with low costing

Extensible

Easy to adapt for different network topologies and attack scenarios

References

1. Assessing Effects of Cyber-Attacks on Smart Grids through Co-Simulation (2025)
2. Towards more realistic co-simulation of cyber-physical energy distribution systems (Oct 2024)
3. Smart Grid Co-Simulation Tools: Review and Cybersecurity Case Study (2024)
4. Intrusion Detection in Smart Power Networks Using Inception-V4 (Jan 2026)
5. Future Cyber-Resilient Renewable and Sustainable Smart Grids (2025)
6. Simulation of multi-stage attack and defense mechanisms in smart grids (Dec 2024)
7. Analysis of the Impact of PV Penetration in Distribution Network (Aug 2025)
8. A hosting capacity-based approach toward distribution system planning (Oct 2024)
9. Application of PandaPower Tool in Evaluating PV Distributed Generation